



Data structure Using JAVA

1. What Is Data structure?
2. Why Data structure.
3. Types of data structure.
4. Types Linear Data structure
5. Array Data Structure.

Data Structures :

In computer terms, a data structure is a specific way to store and organize data in a computer's memory so that this data can be used efficiently later.

Data may be arranged in many different ways, such as the logical or mathematical model for a particular organization of data is termed as a data structure.

1. **Efficiency:** Efficiency of a program depends upon the choice of data structures. For example: suppose, we have some data and we need to perform the search for a particular record. In that case, if we organize our data in an array, we will have to search sequentially element by element. hence, using array may not be very efficient here. There are better data structures which can make the search process efficient like ordered array, binary search tree or hash tables.
2. **Reusability:** Data structures are reusable, i.e. once we have implemented a particular data structure, we can use it at any other place. Implementation of data structures can be compiled into libraries which can be used by different clients.
3. **Abstraction:** Data structure is specified by the ADT which provides a level of abstraction. The client program uses the data structure through interface only, without getting into the implementation details.

1 Linear Data Structures

2. Non-Linear Data Structures.

Linear Data structure means elements combine in the form any specific order. There are two techniques of representing such linear structure within memory.

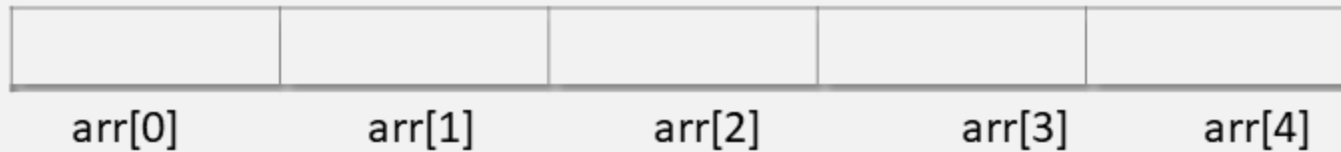
The first way is to provide the linear relationships among all the elements represented using linear memory location. These linear structures are termed as **arrays**.

The second technique is to provide a linear relationship among all the elements represented by using the concept of pointers or links. These linear structures are termed as linked lists.

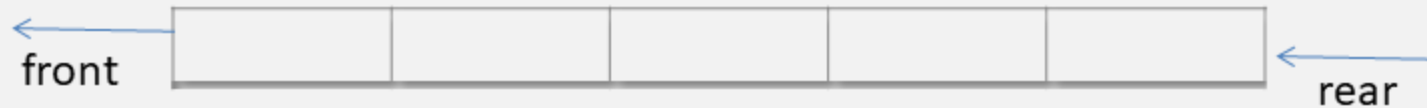
The common examples of the linear data structure are:

- Arrays
- Queues
- Stacks
- Linked lists

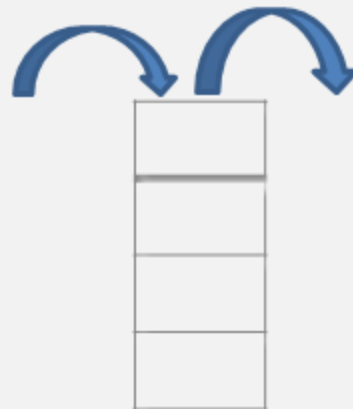
1. Array



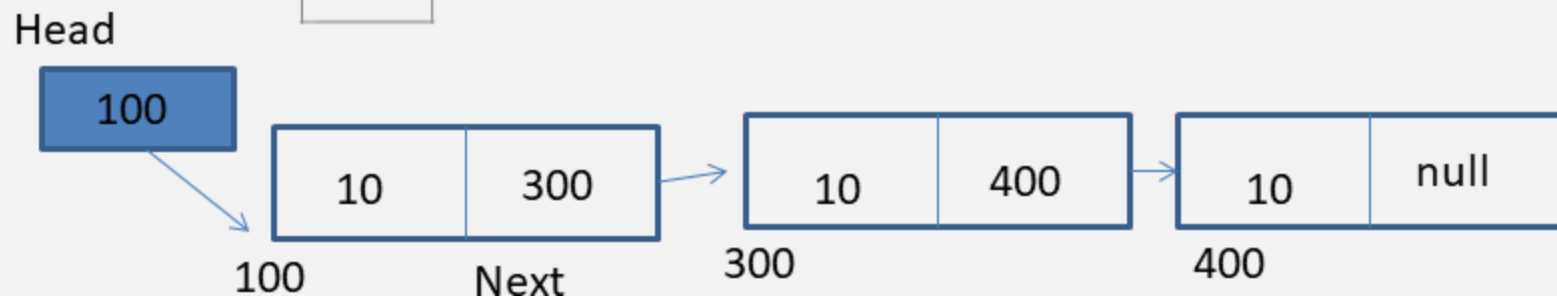
2. Queue



3. Stack



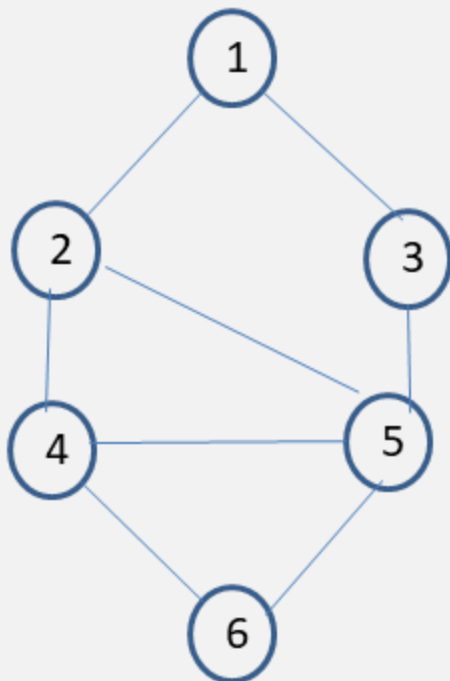
4. Linked List



If data is not in any specific order then we need non linear data structure. This structure is mostly used for representing data that contains a hierarchical relationship among various elements.

Examples of Non-Linear Data Structures are listed below:

- Graphs



- Trees

